

Ergodicity of the projection-preserving switch chain on deduplicated single-sender directed hypergraphs

Abstract

We record the state space, the move set, and the ergodic properties of the Metropolis chain used to sample the exponential tilt $P_\theta(H) \propto e^{\theta\Phi(H)}$ over regroupings of a fixed directed projection. Symmetry of the proposal, aperiodicity, and the reduction to a per-sender problem are proved in full. Irreducibility is proved on the *unrestricted* (row-repetition permitted) state space by reduction to Ryser’s interchange theorem, and on the *deduplicated* state space in the case where every hyperedge has size two, by reduction to the classical switch-connectivity theorem for simple graphs. For the deduplicated space with general size profiles we state the connectivity claim as a conjecture and report exhaustive computational verification; we are not aware of a proof, and we note that the analogous gap is present in the literature this construction builds on.

1 Setup

Definition 1 (Single-sender directed hypergraph). Let V be a finite node set. An *event* is a pair (s, R) with $s \in V$ and $R \subseteq V \setminus \{s\}$, $R \neq \emptyset$; s is the *sender* (tail) and R the *recipient set* (head). The *team* of the event is $N = \{s\} \cup R$ and its *size* is $k = |N|$. A single-sender directed hypergraph is a finite collection H of events.

Following Abuisa et al. [?], we distinguish *edge-ordered* hypergraphs, in which hyperedges carry unique identifiers, from *edge-unordered* ones, in which they do not. We work throughout with the *deduplicated* (or *simple*) case, in which H is a *set* of events: no event is repeated. This is strictly stronger than being edge-unordered, where the edge collection is a multiset whose elements merely lack identifiers.

Definition 2 (Projection and fiber). The *weighted directed projection* of H is the map $\pi_H : V \times V \rightarrow \mathbb{Z}_{\geq 0}$,

$$\pi_H(u, v) = \#\{(s, R) \in H : s = u, v \in R\}.$$

Given an observed H_0 , the *fiber* $\mathcal{F}(H_0)$ is the set of deduplicated single-sender directed hypergraphs H such that

(F1) $\pi_H = \pi_{H_0}$ (identical weighted directed projection), and

(F2) for every sender s , the multiset of sizes of s ’s events in H equals that in H_0 .

Condition (F1) is what makes graph reciprocity constant across the fiber; (F2) is an identification choice, conditioning out the hyperedge-size composition, which mechanically influences every group-reciprocity measure.

Definition 3 (Switch). Let $H \in \mathcal{F}(H_0)$, let s be a sender, and let $E_i \neq E_j$ be two of s 's recipient sets. Let $a \in E_i \setminus E_j$ and $b \in E_j \setminus E_i$. The *switch* $\sigma_{s,i,j,a,b}$ replaces E_i by $E'_i = (E_i \setminus \{a\}) \cup \{b\}$ and E_j by $E'_j = (E_j \setminus \{b\}) \cup \{a\}$, leaving all other events unchanged. The switch is *admissible* if the resulting collection is still deduplicated.

Definition 4 (The chain). Fix $\theta \in \mathbb{R}$ and a statistic $\Phi : \mathcal{F}(H_0) \rightarrow \mathbb{R}$. One step of the chain from H : choose a sender s uniformly from those with at least two events, choose ordered indices i, j uniformly and independently from s 's m_s events, choose a uniformly from $E_i \setminus E_j$ and b uniformly from $E_j \setminus E_i$. If $i = j$, or either difference set is empty, or the switch is inadmissible, remain at H . Otherwise move to $H' = \sigma_{s,i,j,a,b}(H)$ with probability $\min\{1, e^{\theta(\Phi(H') - \Phi(H))}\}$ and remain at H otherwise.

Remark 5. Every failed proposal consumes a step and is a self-loop. Resampling until a valid proposal is found would replace the transition kernel $p(H, \cdot)$ by $p(H, \cdot)/(1 - p_{\text{fail}}(H))$; since p_{fail} is state-dependent, the resulting chain is not symmetric and its stationary law is proportional to $e^{\theta\Phi(H)}(1 - p_{\text{fail}}(H))$ rather than $e^{\theta\Phi(H)}$.

2 Structural lemmas

Lemma 6 (Invariance). *Every switch preserves (F1) and (F2). Consequently the chain never leaves $\mathcal{F}(H_0)$.*

Proof. A switch moves recipient a from E_i to E_j and b from E_j to E_i , both within the events of a single sender s . For $r \notin \{a, b\}$ the count $\pi(s, r)$ is untouched; $\pi(s, a)$ loses one occurrence in E_i and gains one in E_j , and symmetrically for b ; no other sender's events change. Hence (F1). Since $|E'_i| = |E_i|$ and $|E'_j| = |E_j|$, the multiset of sizes is preserved, giving (F2). Admissibility preserves deduplication by definition. $\square$

Lemma 7 (Involution and symmetry). *Write $A = E_i \setminus E_j$, $B = E_j \setminus E_i$, $C = E_i \cap E_j$. After the switch, $A' = E'_i \setminus E'_j = (A \setminus \{a\}) \cup \{b\}$ and $B' = E'_j \setminus E'_i = (B \setminus \{b\}) \cup \{a\}$; in particular $|A'| = |A|$ and $|B'| = |B|$. The map $\sigma_{s,i,j,b,a}$ inverts $\sigma_{s,i,j,a,b}$, and the proposal kernel q satisfies $q(H, H') = q(H', H)$.*

Proof. $E'_i = C \cup (A \setminus \{a\}) \cup \{b\}$ and $E'_j = C \cup (B \setminus \{b\}) \cup \{a\}$. As A, B, C are pairwise disjoint and $a \neq b$, the stated forms of A' and B' follow, and the cardinalities are preserved. Applying the switch with the roles of a and b exchanged restores E_i and E_j . The proposal probability of the forward move is $\frac{1}{|S|} \cdot \frac{1}{m_s^2} \cdot \frac{1}{|A|} \cdot \frac{1}{|B|}$ and of the reverse move $\frac{1}{|S|} \cdot \frac{1}{m_s^2} \cdot \frac{1}{|A'|} \cdot \frac{1}{|B'|}$; these agree. $\square$

Lemma ?? is the analogue in our setting of [?, Theorem 2], which establishes double stochasticity of the DiNgHy transition matrix.

Lemma 8 (Identity switches). *$|A| = |B| = 1$ if and only if the switch exchanges E_i and E_j , i.e. $E'_i = E_j$ and $E'_j = E_i$. In the deduplicated (edge-unordered) state space such a switch is the identity.*

Proof. If $A = \{a\}$ and $B = \{b\}$ then $E_i = C \cup \{a\}$ and $E_j = C \cup \{b\}$, so $E'_i = C \cup \{b\} = E_j$ and $E'_j = C \cup \{a\} = E_i$. Conversely if $E'_i = E_j$ then $(A \setminus \{a\}) \cup \{b\} = B$, and since $b \notin A$ this forces $A = \{a\}$; symmetrically $B = \{b\}$. $\square$

Lemma 9 (Factorisation). $\mathcal{F}(H_0) \cong \prod_s \mathcal{F}_s(H_0)$, where $\mathcal{F}_s$ is the set of deduplicated collections of s 's events satisfying (F1) and (F2) for that sender alone. Every switch acts within a single factor. Hence the chain is irreducible on $\mathcal{F}(H_0)$ if and only if its restriction to $\mathcal{F}_s$ is irreducible for every s .

Proof. Conditions (F1) and (F2) are stated per sender, and by Lemma ?? a switch modifies only the events of one sender. Deduplication also decomposes: two events with different senders are automatically distinct. The product structure and the componentwise action follow, and irreducibility of a product chain in which each step moves one coordinate is equivalent to irreducibility of each coordinate chain. $\square$

By Lemma ?? we may fix a single sender s and drop it from the notation. Let $R_1, \dots, R_n$ enumerate the recipients ever contacted by s , let $c_r = \pi(s, R_r)$, and let $E_1, \dots, E_m$ be s 's events. Encode the state as the $m \times n$ incidence matrix M with $M_{ir} = \mathbf{1}[R_r \in E_i]$. Then (F2) fixes the row sums, (F1) fixes the column sums, and deduplication says the rows of M are pairwise distinct. A switch is exactly the classical *interchange*: choosing rows $i \neq j$ and columns $a \neq b$ with $M_{ia} = M_{jb} = 1$, $M_{ib} = M_{ja} = 0$, and replacing the 2×2 submatrix

$$\begin{pmatrix} 10 \\ 01 \end{pmatrix} \text{ by } \begin{pmatrix} 01 \\ 10 \end{pmatrix}.$$

Lemma 10 (Aperiodicity). *For every $H \in \mathcal{F}(H_0)$, $p(H, H) \geq c$ where $c = \frac{1}{|S|} \sum_{s \in S} \frac{1}{m_s} > 0$ is independent of H . In particular the chain is aperiodic.*

Proof. The event $\{i = j\}$ has probability $1/m_s$ once sender s is chosen, and results in a self-loop. By (F2) the number m_s of events of each sender is invariant along the chain, so c does not depend on the state. $\square$

Remark 11. Lemma ?? shows that making the chain *lazy* — replacing p by $\lambda p + (1 - \lambda)I$ — is unnecessary here: the required uniform lower bound on the holding probability is already available. Laziness preserves the stationary law and reversibility, but multiplies the mixing time by $\Theta(1/\lambda)$, and [?] indeed report designing their proposal to *reduce* self-loops for faster mixing.

3 Irreducibility

3.1 The unrestricted state space

Let $\mathcal{X}$ denote the set of $m \times n$ zero-one matrices with the given row and column sums, *without* the distinct-rows requirement — equivalently, the edge-ordered state space in which a sender may repeat an event.

Theorem 12 (Ryser). *Any two matrices in $\mathcal{X}$ are connected by a finite sequence of interchanges. Consequently the switch chain restricted to $\mathcal{X}$ is irreducible, and together with Lemmas ?? and ?? it is ergodic with stationary law proportional to $e^{\theta\Phi}$.*

Proof. This is Ryser's interchange theorem [?]; see also Brualdi's survey [?]. In the bipartite-graph formulation — rows and columns as the two sides, $M_{ir} = 1$ as an edge — it is the statement that the switch chain on bipartite graphs with prescribed degree sequences is irreducible, as used by Kannan, Tetali and Vempala [?] and by Abuissa et al. [?]. $\square$

3.2 The deduplicated state space

Let $\mathcal{S} \subsetneq \mathcal{X}$ be the subset of matrices with pairwise distinct rows, and let switches be admissible only when they stay in $\mathcal{S}$. Removing states from a connected graph can disconnect it, so Theorem ?? does *not* imply that $\mathcal{S}$ is connected, and the question is genuinely different.

Two special cases are settled.

Proposition 13 (Distinct sizes). *If the row sums $|E_1|, \dots, |E_m|$ are pairwise distinct then $\mathcal{S} = \mathcal{X}$ and the chain is irreducible on $\mathcal{S}$ by Theorem ??.*

Proof. Rows of different sums cannot be equal, so the distinctness constraint is vacuous, and every interchange is admissible. $\square$

Theorem 14 (All events dyadic). *If every event has size $k = 3$, i.e. every row sum equals 2, then the switch chain is irreducible on $\mathcal{S}$.*

Proof. Identify each row, a 2-subset of recipients, with an edge of a graph G on the recipient vertex set; the column sums are the degrees of G and distinctness of rows says exactly that G is simple. An admissible interchange takes two disjoint or sharing edges and rewires them without creating a multi-edge, which is precisely the double edge swap. The theorem is then the classical statement that any two simple graphs with the same degree sequence are connected by double edge swaps, due to Petersen [?] in the regular case and in the general form given by Havel [?], Hakimi [?], Eggleton and Holton [?] and Taylor [?]. See [?] for a modern treatment aimed at sampling. $\square$

For general size profiles we do not have a proof.

Conjecture 15. *For every choice of row sums and column sums, the admissible-interchange graph on $\mathcal{S}$ is connected.*

Computational verification. Conjecture ?? was checked exhaustively on small instances. For each $n \leq 7$ recipients and each size profile drawn from uniform profiles $(k, \dots, k)$ with $k \in \{2, 3, 4\}$ and $m \leq 6$ rows, and mixed profiles combining sizes 2, 3, 4, every fiber was enumerated, grouped by column-sum vector, and the admissible-interchange graph on each fiber with at least two states was tested for connectivity by breadth-first search. Across two independent sweeps this covered roughly 4.3×10^4 non-trivial fibers, the largest containing 2367 states. *No disconnected fiber was found.*

Remark 16. This is evidence, not proof, and the reader should treat Conjecture ?? accordingly. We note that the same gap is present in the closest prior work: Abuissa et al. [?] show that the Markov chain underlying their sampler is *not* irreducible in general, and supply only sufficient conditions for irreducibility; they also observe that no general mixing-time analysis is available even for the dyadic switch chain [?]. A practical mitigation, which we adopt, is to verify empirically that chains started from several distinct states in the same fiber converge to a common distribution of Φ .

4 Consequence

Corollary 17. Assume Conjecture ??, or either of the hypotheses of Proposition ?? and Theorem ?. Then the chain is irreducible (Lemma ??) and aperiodic (Lemma ??) on $\mathcal{F}(H_0)$, its proposal is symmetric (Lemma ??), and hence the Metropolis chain has unique stationary distribution

$$P_\theta(H) = \frac{e^{\theta\Phi(H)}}{\sum_{H' \in \mathcal{F}(H_0)} e^{\theta\Phi(H')}}, \quad H \in \mathcal{F}(H_0).$$

At $\theta = 0$ this is the uniform distribution on $\mathcal{F}(H_0)$, i.e. the projection-preserving null model. Since every $H \in \mathcal{F}(H_0)$ has $\pi_H = \pi_{H_0}$, the reciprocity of the directed projection is constant on $\mathcal{F}(H_0)$ and therefore invariant in θ .

Remark 18. Because the state space is deduplicated, $|E| = \sum_N p_N$, the number of distinct directed hyperedges, is constant on $\mathcal{F}(H_0)$. As $|E|$ is the denominator of the group-reciprocity measures, those measures are then computed against a fixed normalisation for every θ . This fails on the edge-ordered space, where an interchange may render two of a sender’s events identical.

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